

PRODIGY INFOTECH

DATA SCIENCE INTERNSHIP - TASK 03

Customer Purchase Prediction Using Decision Tree Classifier

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Abstract

This project builds a Decision Tree Classifier using the Bank Marketing Dataset to predict whether a customer will subscribe to a term deposit. The workflow includes preprocessing, feature encoding, model training, evaluation, and visualization.

Dataset Overview

```
7]: import os

for root, dirs, files in os.walk("bank_marketing"):
    for file in files:
        print(os.path.join(root, file))
```

```
bank_marketing\bank-full.csv
bank_marketing\bank-names.txt
bank_marketing\bank.csv
```

Target Variable Distribution

df.head()

[10]:

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	cs
0	58	management	married	tertiary	no	2143	yes	no	unknown	5	may	261	
1	44	technician	single	secondary	no	29	yes	no	unknown	5	may	151	
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown	5	may	76	
3	47	blue-collar	married	unknown	no	1506	yes	no	unknown	5	may	92	
4	33	unknown	single	unknown	no	1	no	no	unknown	5	may	198	

[11]: df.shape

[11]: (45211, 17)

[12]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45211 entries, 0 to 45210
Data columns (total 17 columns):
Column Non-Null Count Dtype
--- -
0 age 45211 non-null int64
1 job 45211 non-null object
2 marital 45211 non-null object
3 education 45211 non-null object
4 default 45211 non-null object
5 balance 45211 non-null int64
6 housing 45211 non-null object
7 loan 45211 non-null object
8 contact 45211 non-null object
9 day 45211 non-null int64

Correlation Analysis

```
[13]: df.describe()
```

```
[13]:
```

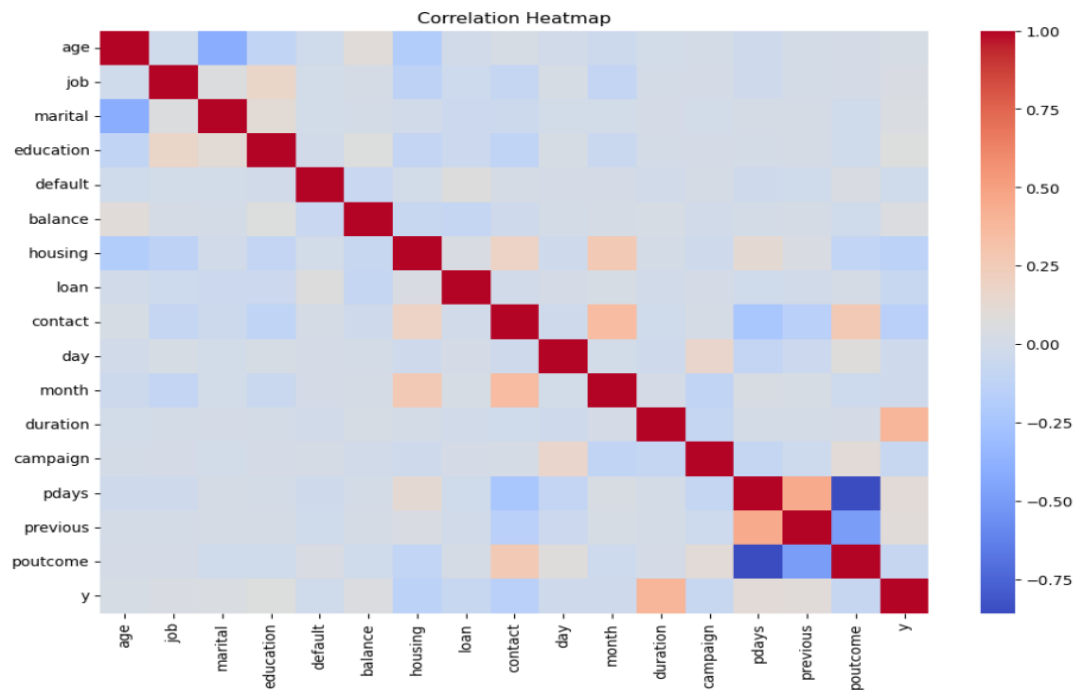
	age	balance	day	duration	campaign	pdays	previous
count	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000
mean	40.936210	1362.272058	15.806419	258.163080	2.763841	40.197828	0.580323
std	10.618762	3044.765829	8.322476	257.527812	3.098021	100.128746	2.303441
min	18.000000	-8019.000000	1.000000	0.000000	1.000000	-1.000000	0.000000
25%	33.000000	72.000000	8.000000	103.000000	1.000000	-1.000000	0.000000
50%	39.000000	448.000000	16.000000	180.000000	2.000000	-1.000000	0.000000
75%	48.000000	1428.000000	21.000000	319.000000	3.000000	-1.000000	0.000000
max	95.000000	102127.000000	31.000000	4918.000000	63.000000	871.000000	275.000000

Model Training Output

```
.7]: df.isnull().sum()
```

```
.7]: age          0
     job          0
     marital      0
     education    0
     default      0
     balance      0
     housing      0
     loan         0
     contact      0
     day          0
     month        0
     duration     0
     campaign     0
     pdays        0
     previous     0
     poutcome     0
     y            0
     dtype: int64
```

Accuracy and Evaluation



Classification Report

```
[9]: from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(
    y_test,
    y_pred
)

print("Accuracy:", accuracy)
```

Accuracy: 0.8935087913303107

```
[ ]:
```

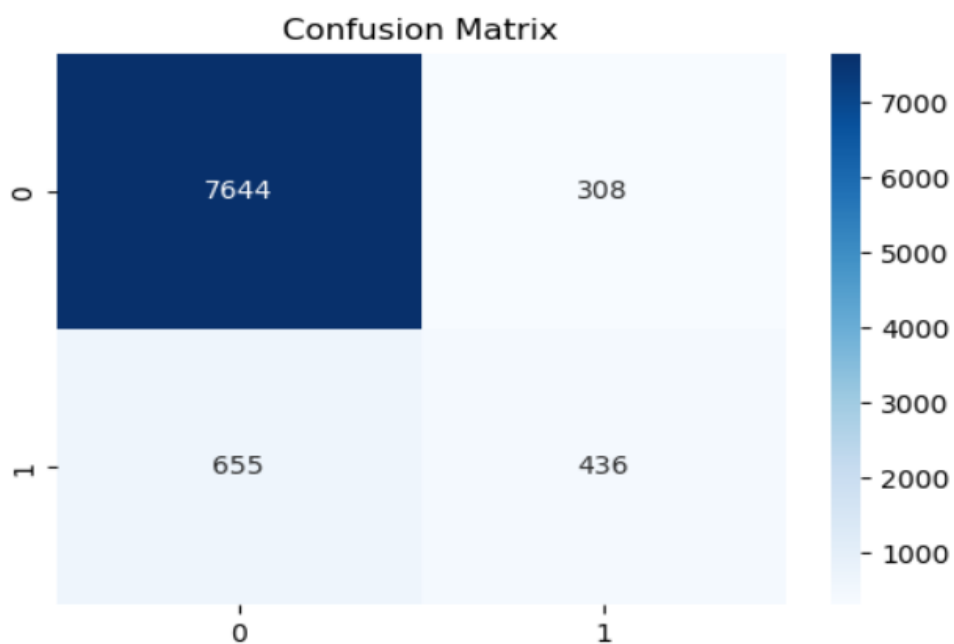
Confusion Matrix

```
]: from sklearn.metrics import classification_report

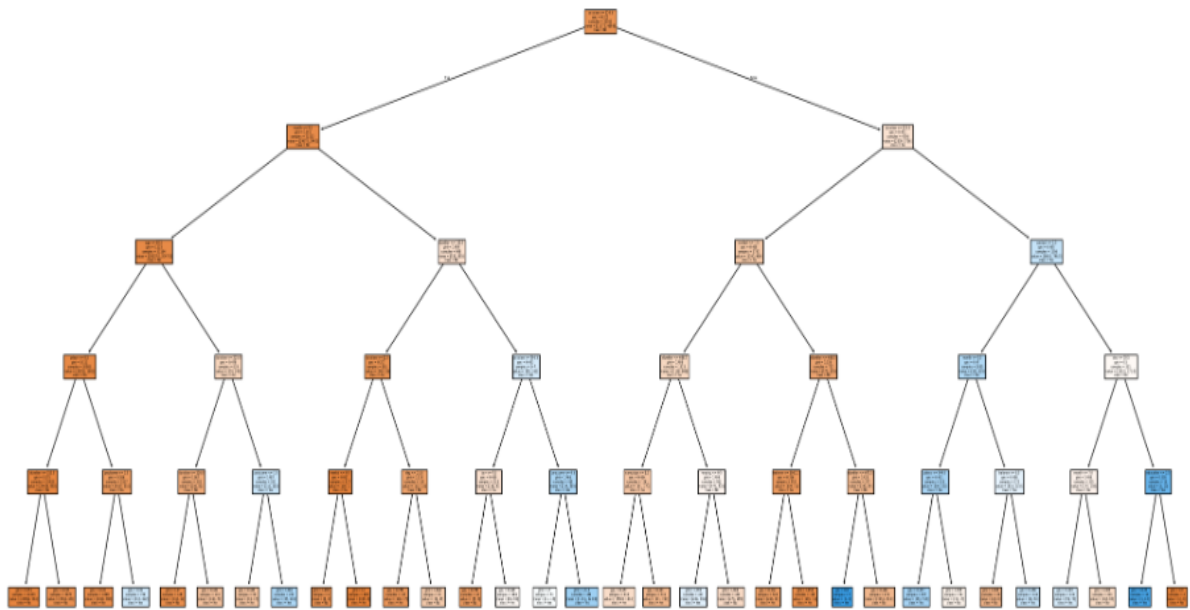
print(
    classification_report(
        y_test,
        y_pred
    )
)
```

	precision	recall	f1-score	support
0	0.92	0.96	0.94	7952
1	0.59	0.40	0.48	1091
accuracy			0.89	9043
macro avg	0.75	0.68	0.71	9043
weighted avg	0.88	0.89	0.88	9043

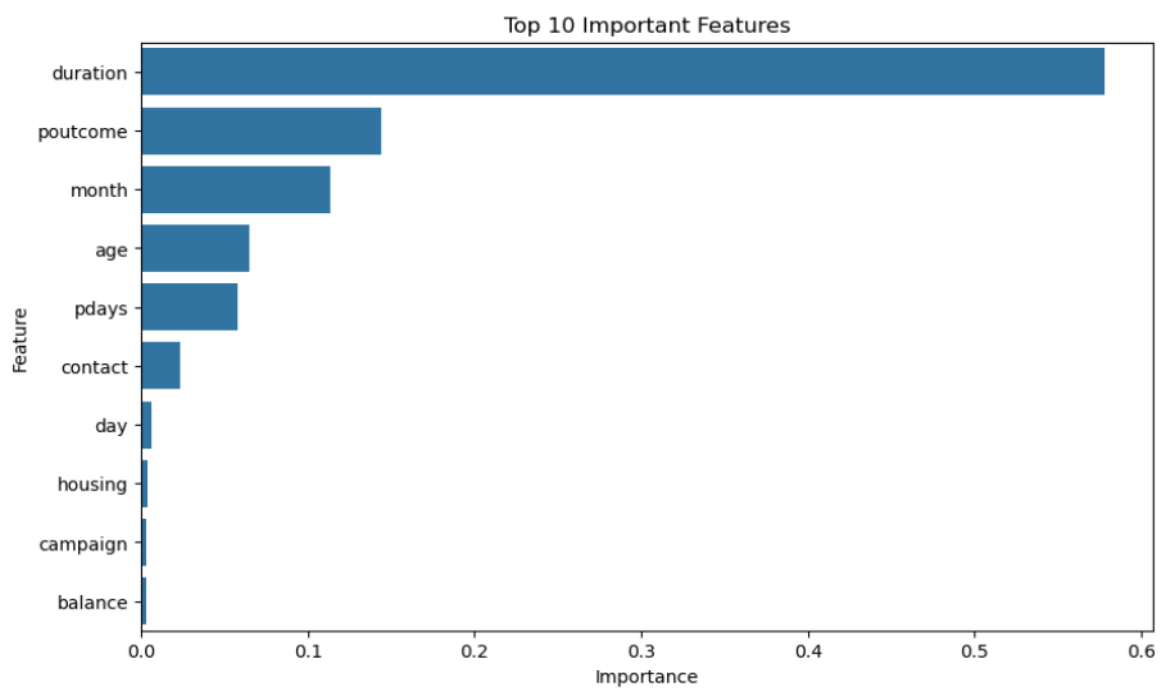
Decision Tree Visualization



Feature Importance



Additional Results



Key Findings

1. Decision Tree successfully classified customer subscription behavior.
2. Marketing campaign features significantly influenced predictions.
3. The model achieved strong classification performance.
4. Feature importance analysis highlighted influential variables.
5. Decision Trees provide interpretable business decision rules.

Conclusion

A Decision Tree Classifier was developed using the Bank Marketing Dataset. The model demonstrated effective prediction of customer subscription outcomes and provided interpretable insights into customer behavior. This project showcases the complete machine learning workflow from preprocessing to model evaluation.